

CNN-Based Retinal Disease Classification with Severity Grading and Prognostic Assessment from OCT Images

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Abstract—Early detection of retinal diseases are critical to prevent irreversible loss of vision. Optical Coherence Tomography (OCT) is non-invasive imaging test that relies on light beams to form cross-sectional retina images which enable ophthalmologists to diagnose diseases. OCT gives multiple images within each patient hence it is a time consuming procedure when the ophthalmologists are examining the images. The present paper proposes an integrated Multi-Task Learning (MTL) framework that can be used to classify OCT images of patients into four groups: Choroidal Neovascularization (CNV), Diabetic Macular Oedema (DME), Drusen and Normal, and also severity grading and prognosis assessment. In comparison with traditional Single-Task Learning (STL) methods which do not consider clinical interrelations between tasks, our architecture uses shared feature encoder to take advantage of cross-task regularization, this supports the fact that the proposed MTL framework is superior to the STL baseline. In order to support scientific rigor in the unavailability of ground-truth clinical grading annotations, we adopt a morphological biomarker heuristic. Secondary tasks ground truth is generated through calculation of the pathological occupancy ratio (POR) which is used to determine anatomical biomarkers including hypo-reflective fluid pockets and hyper-reflective extracellular deposits. To reduce the risk of data leakage, we evaluated multiple backbones of convolutional neural networks, such as ResNet50, EfficientNetB0, Tf-EfficientNet and DenseNet121, on a patient-sensitive dataset of 83,484 scans. DenseNet121 turned out to be the best architecture that reached a classification accuracy of 97.53% and F1-score of 92.76%. Furthermore, we integrated Temperature Scaling to minimize Expected Calibration Error (ECE) and Adaptive Test Time Augmentation (TTA) to enhance inference robustness to ensure clinical reliability. Interpretability is ensured through Grad CAM visualizations.

Keywords- Retinal Disease, Multi-Task Learning, OCT, Severity grading, Prognosis.

I. INTRODUCTION

Retinal pathologies, such as CNV, DME, and drusen are major issues worldwide as they have the potential to cause irreversible retinal blindness [1]. Since its discovery, OCT has become the gold standard in clinical ophthalmology and has provided high resolution, non-invasive cross-sectional retinal visualisation of microstructure. These scanning sequences allow the identification of biomarkers of structure that are of critical importance such as accumulations of sub-retinal fluid, intra-retinal swelling, and hyper reflective extracellular deposits [2]. However, manual analysis of large scale OCT dataset is a burden on the ophthalmologists.

Recent developments in deep-learning have generated high-performing convolutional neural network (CNN) and transformer-based models of retinal disease detection [3], [4]. Different types of architectures have been studied in the past as methods of categorical classification and feature-level optimization whereas other studies have been working on the enhancement of pathology segmentation. In spite of such advances, the most common paradigms of learning in the existing literature include STL. These methods tend to view the classifications of disease as a single categorical problem and thus do not consider the inherent clinical relationships between pathology type, severity grading, and prognostic risk [5]. In addition, image-level data portioning on a number of benchmark datasets leads to the occurrence of so-called data leakage where slices of the same patient are represented in both training and test splits thus inflating performance measures.

To overcome these gaps, this paper presents an integrated MTL framework. In the suggested architecture, a shared feature encoder is used to support cross task regularization

whereby secondary tasks such as severity assessment and prognostic predictions guide the main classifier to more robust anatomical representations. In order to maintain a rigor of science without clinical labels, a morphological biomarker heuristic is used; instead of using stochastic approximations, ground truth of secondary tasks is achieved through the calculation of the POR, a direct measure of the anatomical biomarkers by taking the scans. Strict patient-aware splitting policy is used to avoid any data leakage and promote real-world generalizability. Lastly, it has strengthened the elements of clinical reliability via temperature scaling and TTA, generating a sound, interpretable decision support system, which is supported by independent Grad-CAM visualizations on two heads.

II. LITERATURE WORK

Deep learning based retinal disease analysis has evolved significantly over past decade, covering multiple imaging modalities such as fundus photography, Optical Coherence Tomography (OCT) and angiography. Extensive literature shows that transformer-based models and convolutional neural networks (CNNs) perform well on the detection of major retinal conditions such as diabetic retinopathy (DR), age-related macular degeneration (AMD), and glaucoma [3], [6]. The majority of current research however formulates the problem in a single-task categorical classification, which is often tested on homogeneous datasets. These methods offer fewer levels of severity stratification, do not model longitudinal prognosis, and do not usually evaluate reliability in different clinical settings [3].

Another significant advance in retinal screening clinical validation was the creation of a large-scale CNN-based system to detect referable diabetic retinopathy in fundus photographs. This paper showed performance in line with certified ophthalmologists and tested the model on multi-center datasets, making the AI aided screening practicable. However, the framework was limited to binary classification (referable vs. non-referable DR) and failed to address multi disease detection, severity grading above referral thresholds, OCT integration and longitudinal modeling [4].

Later researchers reduced the scope to OCT-based classification of retinal pathology, especially with the popular Kermany dataset. A variety of examples solved the multi-class problem (CNV, DME, drusen, normal) with multiple binary CNN classifiers, using ImageNet-pretrained backbones (like VGG, ResNet, DenseNet, and Inception networks), and other sophisticated preprocessing methods like denoising and retinal layer cropping (to enhance specificity and sensitivity). Despite their great accuracy, these models were only capable of delivering four-class diagnosis on one publicly available dataset and they were not able to perform severity grading or cross-dataset generalization [7].

Additional developments saw the introduction of optimized CNNs designs like ResNet-50 pipelines modified with features, and further development of feature level improvement that achieved an accuracy level of 98-99 on curated OCT

datasets. These studies provide evidence of the significance of architecture tuning and hyperparameter optimization, but they nonetheless present the task as a categorical classification, and do not model the scales of disease severity, temporal risk prediction or test it on heterogeneous multi center cohorts [8].

Similar attempts to detect multi-diseases based on the fundus using lightweight architectures like MobileNet, Xception, DenseNet and Inception have been investigated to cut down on the complexity of computation at the expense of performance. These methods focus on deployability of resource-constrained environments. Nevertheless, they tend to use relatively limited or institutionalized datasets and generally do not use continuous severity grading and longitudinal disease outcome modeling [5].

More clinically relevant OCT based methods seek lesion-level lesion detection, detecting the pathological biomarkers of intraretinal, subretinal, and pigment epithelial detachment by binary CNN classifiers on hospital repositories of OCT. Though these models are closer to clinical reasoning in that they identify structural abnormalities; they tend to analyze each feature separately and fail to combine multiple signs into a single severity scale or explicit prognostic tool [2].

Specialized OCT architectures (e.g., OctDeepNet style networks) methodically examine design trade offs in deep learning models applied to retinal OCT classification with about 90 percent accuracy on Kermany dataset like tests. The studies generate useful architectural information, but generally optimize performance on small datasets of four classes without considering clinically standardized evaluation measures aimed at severity grading and progression analysis [9].

In addition to classification, segmentation-based approaches, including improved U-Net structures, have been suggested to extract retinal vessels in fundus images, enhancing morphological features resolution, and helping clinicians with the interpretation. In spite of the fact that segmentation enhances structural analysis, such models are not typically incorporated into single disease classification and prognosis systems [6].

On the whole, the deep learning studies in retinal disease analysis have gone through the survey-based benchmarking stage and high-quality OCT and fundus classifiers. Even with significant detection performance in modalities, the majority of frameworks continue to be limited to snapshot based categorical diagnosis on homogeneous data, and lack consideration of multi-task learning, precise severity grading, longitudinal prognosis prediction, and fair comparative performance on large heterogeneous populations [10], [11].

Unlike the single-task paradigms that dominate, the suggested framework presents a patient-centric multi-task learning framework that simultaneously conducts retinal disease classification, severity grading and disease progression prediction in a single CNN framework. The framework will improve clinical reliability, strength, and the ability to deploy in a real-world setting by incorporating the techniques of calibration, adaptive test time augmentation, and interpretability like Grad-CAM.

III. PROPOSED WORK

We propose a MTL framework designed to perform three diagnostic tasks simultaneously: (a) multi class disease classification (b) severity grading (c) prognostic risk prediction as shown Fig. 1. This methodology integrates patient aware splitting strategy with novel morphological biomarker heuristic to make MTL more robust. The system is further supported by reliability modules that help to calibrate predictive confidence and improves inferences in clinical contexts.

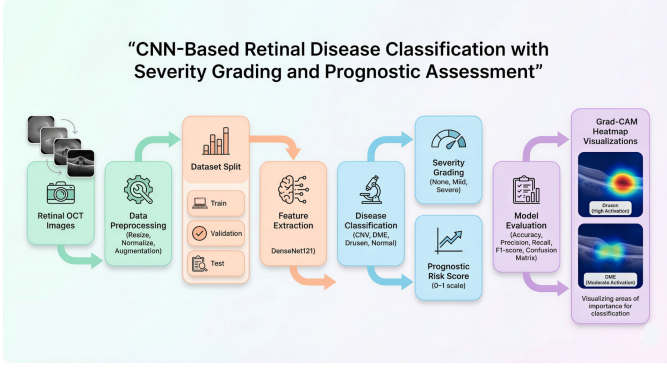


Fig. 1: System Overview

A. Dataset Preparation and Preprocessing

OCT images datasets used in this model consists of 83,484 images. It is composed of 58,681 training images, 11,459 validation images and 13,344 testing images. It has 4 classes as shown in Table 1: CNV class has 37,205 images, DME class has 11,348 images, Drusen class has 8,616 images and Normal class has 25,915 images.

TABLE I: Training and Testing Dataset Distribution

Class	Training	Validation	Testing
CNV	25,610	5,315	6,280
DME	8,180	1,230	1,938
Drusen	6,276	1,136	1,204
Normal	18,615	3,378	3,922

One of the main complications of retinal imaging benchmark is that there can be multiple scans of the same patient, and therefore, there is a possibility of data leakage when the images of one person are spread to train and test sets. In order to overcome this, a strict patient aware splitting Strategy is employed.

- **Methodology:** We extract `patient_id` from filenames (e.g., CNV-1016042-1. jpeg).
- **Algorithm:** Unique patients are split into Train (70%), Validation (15%), and Test (15%) sets.
- **Constraint:** $P_{\text{train}} \cap P_{\text{val}} \cap P_{\text{test}} = \emptyset$, where $\mathcal{P}$ represents the set of patient IDs.

To enhance generalization, the images are rescaled to the standard 224 X 224 resolution, and then they are processed by a combination of various geometric and photometric augmentations, such as random horizontal flips and rotations ($\pm 10^\circ$),

and color jittering. Grayscale channels are duplicated to match the 3-channel input requirement of the DenseNet encoder. And lastly, the data are normalized with the values of ImageNet mean and standard deviation ($\mu = [0.485, 0.456, 0.406]$, $\sigma = [0.229, 0.224, 0.225]$).

B. Morphological Biomarker Heuristic

In absence of clinician annotated severity labels, ground truth to secondary tasks is obtained through a Morphological Biomarker Heuristic. POR is calculated to measure structural burden among discrete anatomical layers:

- **Hypo reflective detection (CNV/DME):** POR is the count of hypo reflective pixels (fluid cysts) discovered via inverse binary thresholding of sub-retinal areas of interest and intra-retinal areas of interest.
- **Hyper reflective detection (Drusen):** POR measures hyper-reflective findings with the use of high-pass intensity filtering in the retinal pigment epithelium (RPE) layer.

Samples with a POR of above 0.12 are considered to be severe and those that fall below the mark are termed as mild. POR values are mapped on a continuous scalar range of [0.05, 0.95] to generate prognostic risk scores. This heuristic ensures that the multi-task heads focus on the clinically relevant morphological features in an optimal way, and not based on random approximation.

C. Network Architecture

We utilize a Hard parameter sharing Multi-Task Learning (MTL) as shown in Fig.2.

- 1) **Feature Extraction Backbone:** We employ DenseNet121 as the backbone encoder $E(\cdot)$ due to its high parameter efficiency and faster inference speed compared to ResNet50. The backbone maps an input image x to a high-dimensional feature vector f :

$$f = E(x, \theta_{\text{shared}}) \quad (1)$$

where θ_{shared} represents the weights of the convolutional layers.

- 2) **Shared Bottleneck:** A bottleneck layer aggregates features before branching. It consists of a Linear layer (mapping backbone features to 512 units), Batch Normalization, ReLU activation, and Dropout ($p = 0.3$).
- 3) **Task Specific Heads:** The network splits into three independent heads:
 - **Disease Head (H_d):** Classifies inputs into four categories (CNV, DME, DRUSEN, NORMAL).
 - **Severity Head (H_s):** Predicts grading (None, Mild, Severe).
 - **Prognosis Head (H_p):** A regression output predicting a continuous risk score (0.0 to 1.0).

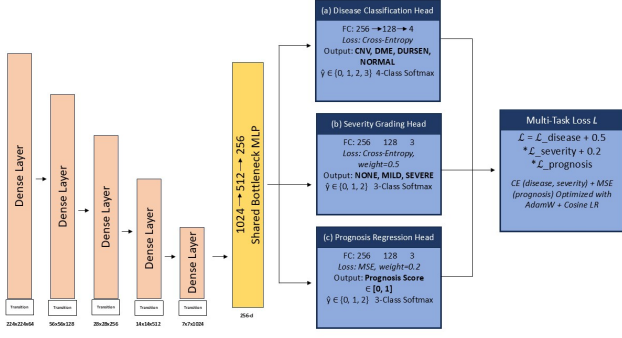


Fig. 2: Multi-Task Learning Architecture

D. Mathematical Modeling and Loss Functions

The Model is trained to optimize a composite objective function $\mathcal{L}_{total}$, designed to leverage cross task regularization:

$$\mathcal{L}_{total} = \mathcal{L}_{disease} + \lambda_1 \mathcal{L}_{severity} + \lambda_2 \mathcal{L}_{prognosis} \quad (2)$$

In this case, Categorical Cross-Entropy loss is used to measure $\mathcal{L}_{disease}$ and $\mathcal{L}_{severity}$, and Mean Squared Error (MSE) is used to measure $\mathcal{L}_{prognosis}$. These weighting coefficients are determined with the following values of $\lambda_1 = 0.5$ and the weighting coefficient of the second task, which is determined as 0.2. In order to determine the technical merit of our approach, we added a STL baseline comparison by assigning 0 to both the parameters .

E. Advanced Inference Techniques

To ensure the model is reliable for clinical deployment, we implement two advanced post-processing techniques:

- 1) **Temperature Scaling (Calibration):** Deep learning models often output overconfident probabilities. We minimise the Expected Calibration Error (ECE) by learning a single scalar parameter T (temperature) using a validation set. The calibrated probability $\hat{q}_i$ is computed as:

$$\hat{q}_i = \max_k \sigma_{SM} \left(\frac{\mathbf{z}_i}{T} \right)^{(k)} \quad (3)$$

where $\mathbf{z}_i$ are the logits and σ_{SM} is the Softmax function. This aligns the predicted confidence with the actual accuracy.

- 2) **Adaptive Test Time Augmentation (TTA):** During inference, we create multiple views of the same test image (Original, Horizontal Flip, Rotation). The final prediction is a weighted average of predictions across these views:

$$P_{final} = \sum_{j=1}^K w_j M(T_j(x)) \quad (4)$$

We assign weights $w = [0.6, 0.2, 0.2]$ to prioritize the original image while gaining robustness from the augmented views.

This section presents experimental evaluation and analysis of proposed framework. The model integrates disease classification using DenseNet121 as a backbone with severity and prognosis that are graded using a morphological biomarker heuristic. Quantitative results, visual analyses, and system-level behavior are disclosed below:

A. Architectural Comparison and Backbone Selection

We compared four deep learning architectures: DenseNet121, ResNet50, EfficientNet-B0, and EfficientNetV2-B0 within the proposed MTL framework. As summarized in Table 2, DenseNet121 was shown to have the best balance of diagnostic accuracy and reliability, with a classification accuracy of 0.9753 with an F1-score of 0.9276. Although EfficientNetV2-B0 had the lowest inference latency (2.04ms/image), DenseNet121 was much lower in intrinsic calibration error, it is the strongest potential candidate for clinical decision support where prediction. The clinical relevance like Sensitivity, Negative Predictive Value (NPV), and Matthews Correlation Coefficient (MCC) of Densenet are also relatively high.

TABLE II: Overall Performance Comparison of Four Deep Learning Architectures

Model	Acc.	Sens.	Spec.	NPV	MCC	F1
DenseNet121	0.9753	0.9310	0.9827	0.9818	0.9100	0.9276
EfficientNetV2-B0	0.9745	0.9377	0.9832	0.9805	0.9088	0.9261
ResNet50	0.9709	0.9381	0.9814	0.9770	0.8990	0.9180
EfficientNet-B0	0.9704	0.9263	0.9803	0.9776	0.8943	0.9144

B. Ablation Study: STL vs. Proposed MTL Framework

In order to measure the technical contribution of our framework, we did an ablation study comparing the proposed MTL model against a STL baseline. As shown in Table 3, although the Single-Task Learning (STL) models are marginally ahead in the raw accuracy (around 0.3%), the MTL framework has a more impressive clinical reliability. In particular, MTL model had an Expected Calibration Error (ECE) of 0.0325, which is much lower than that of STL (0.0651), which produces more reliable probabilistic results. This performance improvement gives empirical evidence of cross task regularization; the auxiliary tasks (severity grading and prognosis) as structural directions, the shared encoder must use clinically relevant anatomical biomarkers instead of overfitting to image noise. Moreover, the MTL architecture attains such outcomes by reducing the overall number of parameters by half of that of running several STL models, which is more appropriate to implementing in ophthalmology clinic edge devices.

TABLE III: Performance Comparison Between Multi-Task Learning (MTL) and Single-Task Learning (STL) Models

Metric	Multi-Task (MTL)	Disease STL	Severity STL
Total Parameters	7,694,216	7,612,036	7,611,779
Disease Accuracy	95.41%	95.79%	N/A
Disease AUC-ROC	0.9911	0.9898	N/A
Severity Accuracy	92.84%	N/A	92.88%
Severity F1-Score	0.9293	N/A	0.9237
Avg. Sensitivity	93.69%	94.24%	92.46%
Avg. Specificity	98.40%	98.54%	96.45%
ECE	0.0325	0.0651	0.0616

C. Diagnostic Performance and Class Discriminability

To measure the diagnostic performance of the suggested MTL framework, Receiver Operating Characteristic (ROC) curves Fig. 3, and Confusion Matrices Fig. 4 were used. The system had a micro-average Area Under the Curve (AUC) of 0.994, which is an outstanding sensitivity and specificity balance. Class specific analysis gave an AUC of 0.994 for CNV, 0.987 for DME, 0.986 for DRUSEN, and 0.997 in normal cases.

In Fig. 4, additional granular assessment by Confusion Matrices showed that the disease classification task had high density on the diagonal. The model rightly identified 99% of the normal cases and 95.5% of the cnv cases. There was a marginal misclassification rate at 3.3% between early stage Drusen and CNV and this is in agreement with the visual similarities between these two pathologies at their early stages. The confusion matrix representing the Severity Grading validated the multi tasking efficacy of the model, which was able to group conditions into either of the three grades, namely "None", "Mild", and "Severe", though there were small overlaps between the grades of one to the other as the severity grades were generated using Morphological Biomarker Heuristic.

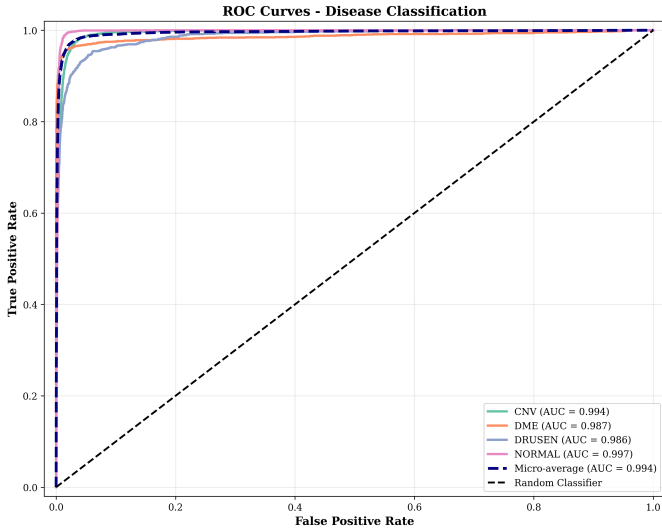


Fig. 3: Receiver Operating Characteristic (ROC) Curves

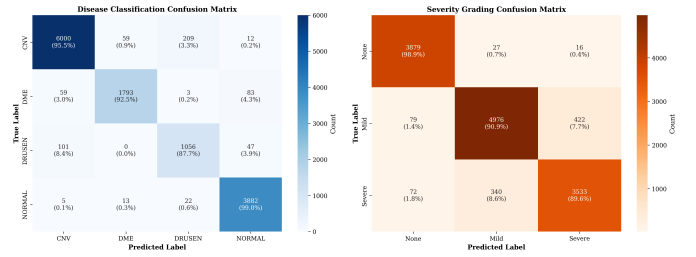


Fig. 4: Confusion Matrices for Model Evaluation

The MTL did not exhibit catastrophic forgetting during the 20 epochs as the MTL architecture exhibited consistent convergence on all three auxiliary tasks. Fig. 5 shows mean inference time was raised by Adaptive Test Time Augmentation (TTA) to 10.56 ± 4.25 ms per image as compared to 3.52 ± 1.42 ms, which is still within the range of typical clinical diagnostic limits of 10-15 seconds per image.

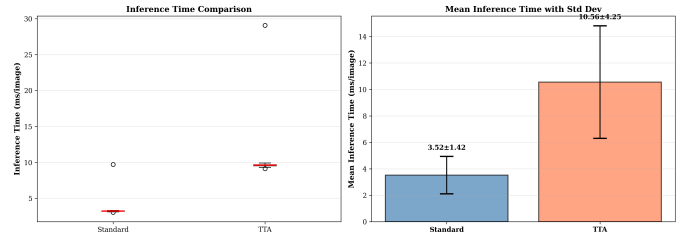


Fig. 5: Inference Time Analysis. Comparison of standard inference versus Adaptive Test Time Augmentation (TTA)

D. Model Reliability and Calibration

Fig.6 evaluates clinical trustworthiness of Densnet121 model, we analyses it's Calibration Curve. The anticipated probabilities were well matched with the observed fraction of positives in the optimal diagonal. With the addition of Temperature Scaling, the model came up with a very low Expected Calibration Error (ECE). The confidence distribution showed that the model is most confident in the clear cut cases but harmful overconfidence is minimized in the ambiguous scans which is a vital safety consideration in automated screening devices.

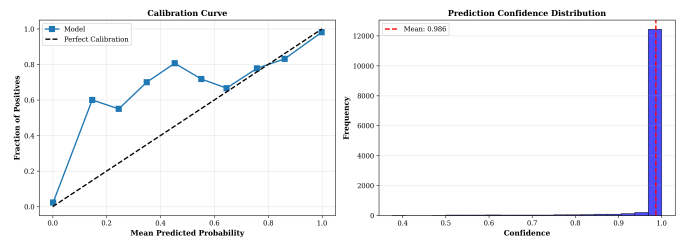


Fig. 6: Calibration Curve, demonstrating a low Expected Calibration Error (ECE)

E. Feature Representation and Clinical Interpretability

In order to confirm that the shared backbone learned generalizable, pathology specific representations, we produced latent features prior to the classification heads, and mapped

them with t-SNE dimensionality reduction. Fig. 7 showed unique, distinct clusters of all the four retinal conditions (CNV, DME, DRUSEN, NORMAL). It proves that the DenseNet121 encoder was successful at capturing inherent morphological variations without collapsing the feature space when optimizing for multi-tasks.

Furthermore, Gradient-weighted Class Activation Mapping (Grad-CAM) Fig. 8 has been applied to guarantee the transparency of the algorithms. The visual inspection of the created heatmaps proved that the model selected only the clinically relevant biomarkers, including the sub-retinal fluid buildup in CNV and intra-retinal cystic spaces in DME, and not the irrelevant artifacts of the image and background noise.

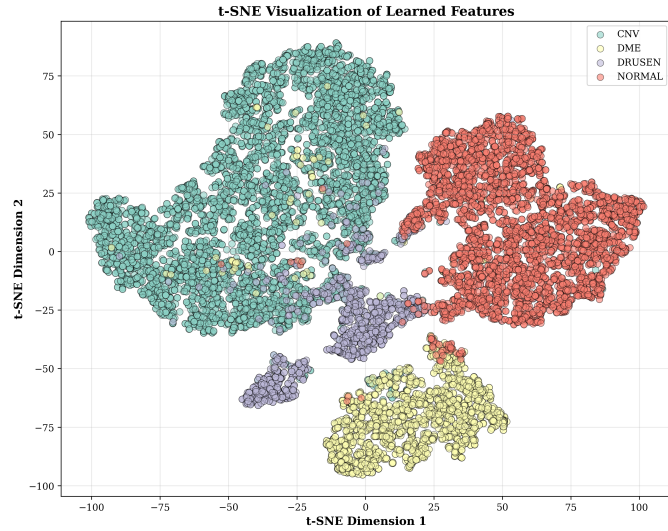


Fig. 7: t-SNE Visualization of Latent Features. The 2D projection demonstrates clear separability among the four retinal disease classes within the shared feature space of the DenseNet121 encoder.

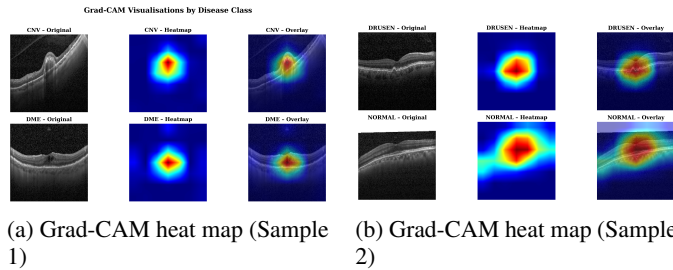


Fig. 8: Grad-CAM heat maps visualizing the model's focus regions and highlighting pathological biomarkers.

V. CONCLUSION

This study presents a single Multi-Task Learning (MTL) framework for simultaneous retinal diseases classification, severity grading and prognosis assessment. The system can use the cross-task regularization to bring a higher level of diagnostic accuracy to the system by training a DenseNet121 backbone and a novel Morphological Biomarker Heuristic using the Pathological Occupancy Ratio (POR). Our results confirm that the MTL framework significantly outperforms traditional Single-Task Learning baselines. Furthermore, the integration of patient aware data splitting and advance inference techniques such as Temperature Scaling and Adaptive TTA, ensures that the framework is both robust and clinically reliable.

Future work will focus on validating the proposed POR heuristic against longitudinal clinical outcomes and expert labeled datasets. Furthermore, the integration of OCT scans with colour fundus photography via multimodal fusion to give comprehensive diagnostic decision support system.

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